

Invertebrate facts

Tiny animals without backbones first appeared more than 600 million years ago. These early invertebrates lived in water, and many still do. Today, the diversity of invertebrates found throughout the world is staggering, from squids and starfish to worms and spiders.

INVERTEBRATE TYPES

There are around 35 main groups of species in the animal kingdom. Just one of these groups, the vertebrates, contains all the fish, amphibians, reptiles, birds, and mammals. The other 34 groups are invertebrates—animals without an internal, jointed skeleton. Six of the main invertebrate groups are shown here.



Sponges

These primitive ocean organisms cannot move, and gather food by filtering it from the water.



Cnidarians

From jellyfish and anemones to corals, these sea creatures all have stinging tentacles to catch small prey.



Echinoderms

With their "spiny skin," these marine animals include starfish, sea cucumbers, and sea urchins.



Mollusks

From slugs to squids, mollusks live in damp habitats or in the sea. Many have a hard shell.



Worms

Found in water and on land, some—such as earthworms—are made up of many identical, soft-skinned segments.



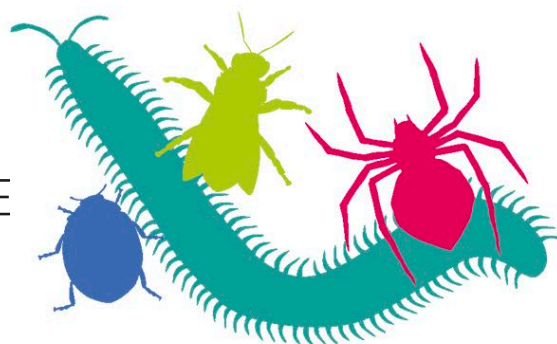
Arthropods

With their tough outer skeletons and jointed legs, arthropods include insects, spiders, and crabs.

ALMOST

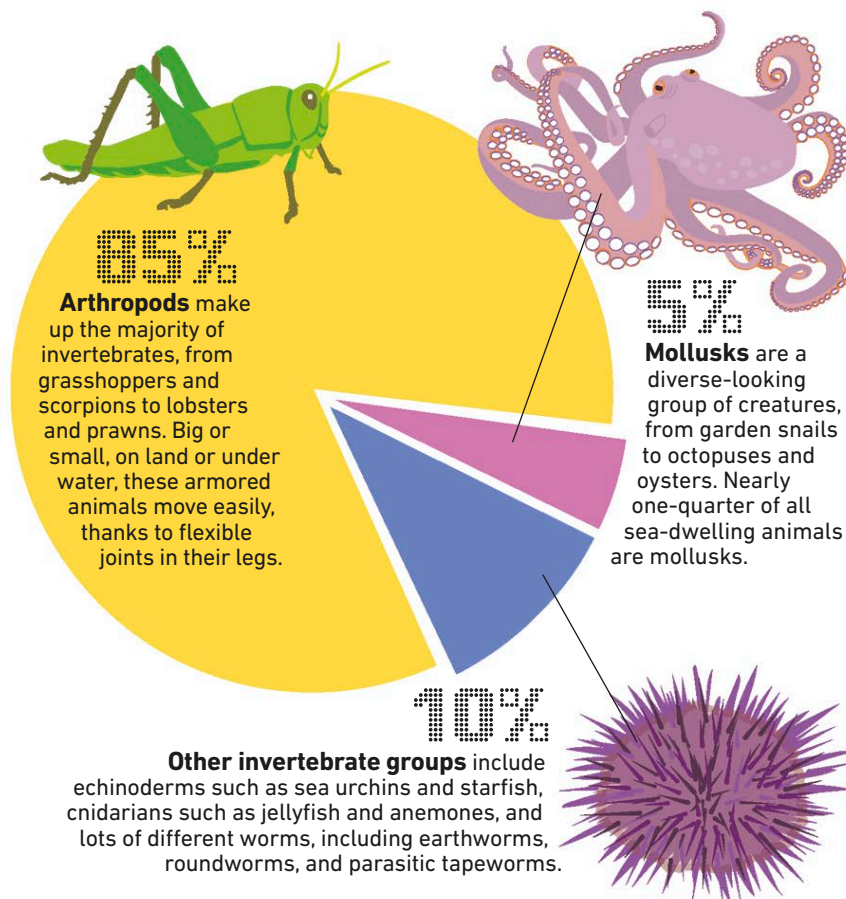
97%

OF ALL ANIMALS ARE
INVERTEBRATES.



INVERTEBRATE NUMBERS

There are approximately 1.3 million known invertebrate species, but there could be many millions more. The vast majority of invertebrates belong to two groups: arthropods and mollusks.



EXTREME HABITATS

Some invertebrates can withstand—and even thrive—in incredibly hostile conditions, from barren, icy Antarctica to vast, unexplored regions thousands of feet below the ocean's surface.



Antarctic midges are insects that measure only $\frac{3}{16}$ in (1 cm), yet are the largest native land animal in Antarctica. They live at temperatures of 5°F (-15°C), spending nine months of the year frozen solid.



Tube worms, a type of marine segmented worm, live on the Pacific Ocean seafloor near hydrothermal vents—volcanic areas where sections of Earth's crust are moving apart. They grow up to 10 ft (3 m).